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1 MAPRO Results with Updated Genome Database (16th of June 2026)

It took almost 20 hours to finish the full analysis. (from jun 16 04.55 pm to jun 17 02.48 pm)

1.1 Old Results

Starting with 120 organisms, 110 of them had at least one available target and neighbor genome assembled to the “complete” level. Of those, 96 had at least one marker region shared across all targets and absent from all neighbors, and I could design a primer pair for 89 of them. Finally, although all of the primers could be scored, it was only possible to correct the primer score for 82 of them. For six primer pairs, the type strain could not be found in the local nt database and the score could not be corrected. To correct the primer score for *Bordetella pertussis* the type strain accession was changed from “BX470248.1” to “CP039022.1”, the complete genome accession present in our local nt database.(taken from Beatriz’s thesis)

1.2 Dataset Summary

Metric	Value
Total taxa analyzed	120 (was 120 in the old results)
Taxa producing phylogenetic trees	97
Taxa producing markers	93 (was 96)
Taxa producing primers	74 (was 89)

1.3 Pipeline Outcomes

1.3.1 Taxa having primers.fasta

afa afi ahy ame ban2 bap bbi bdi bfa bfr blo bma bpa bpe bps bth1 bth2
cac cau cbo1 cbo2 cbu cdi cgl cje cpn cte ctr1 cvi1 cvi2 dra dtu dvu ecl eco2
efa1 efa2 fps gsu gvi hin kvu lac lin lpn lsa mab mci mfl mle msm mth ngo
nme psy1 psy2 rba rru sag sco sdy sep sme2 son1 spy ssa tac tma tth tye
vpa xca yen ype

1.3.2 Taxa that have markers.fasta and don't have primers.fasta (some of them are empty)

The ones that contain markers have too many Ns in the sequencesq

```
cd analysis
for a in */; do
    [ -f "$a/markers.fasta" ] && [ ! -f "$a/primers.fasta" ] && \
    { echo -n "$a "; grep -c '^>' "$a/markers.fasta"; }
done
```

```
api/ 8 bbr/ 0 bbu/ 0 bce/ 0 eco1/ 0 eco3/ 0 eco4/ 0 eco5/ 0 gva/ 0 kae/
0 lmo/ 13 mtu2/ 0 pma/ 0 ppu/ 3 sau/ 1 sen2/ 0 sfl/ 1 ssu/ 45 tde/ 0
```

1.3.3 Taxa that don't have markers (27)

```
for a in */; do
    [ ! -f "$a/markers.fasta" ] && [ ! -f "$a/primers.fasta" ] && echo "${a%/*}"
done | sort
```

```
aae ban1 bsu ctr2 eco6 fnu ftu hpy kpn lla lpa lpl mmy mpn mtu1 pae
rpr rsp sen1 sfr sme1 smi smu spn2 sru tel vch
```

```
for a in $(ls -d */); do
    echo -n "$a "
    cres $a/all/*
done
```

1.3.5 Penalty Scores of the Taxa

```
for d in ???/; do
    [ -f "$d/primers.fasta" ] && \
        echo "$(basename "$d")" $(grep -m1 'penalty:' "$d/primers.fasta" | sed 's/.*penalty: //')
done | sort
```

1.3.6 split percentages for each tax

- 3

all results

```
cd analysis
for d in */; do
    if [ -f "$d/all.nwk" ]; then
        split=$(fintac "$d/all.nwk" | tail -n +2 | sort -n | awk '{print $5}')
        echo "$d $split"
    fi
done | sort -nk2
```

taxa that have 100% split

```
for d in */; do
    if [ -f "$d/all.nwk" ]; then
        split=$(fintac "$d/all.nwk" | tail -n +2 | sort -n | awk '{print $5}')

        if [ "$split" = "100.00" ]; then
            echo "$d $split"
        fi
    fi
done | wc -l
```

51 of them had 100% split

worst 3 taxa:

- vch %66.67
- psy1 %80.00
- bce %82.47

"Generally speaking, taxa with more genomes had a better split, but there are some exceptions. One of these exceptions is *Salmonella enterica*, marked in red in Figure 3.19. Despite having 1201 genomes, the split score is relatively low, 93.3%, compared to the other taxa." Now we do not even have markers

1.3.7 markers

```
find . -name markers.fasta | wc -l
```

93 taxa have marker
total markers

```

for d in */; do
    [ -f "$d/markers.fasta" ] || continue
    grep -c '^>' "$d/markers.fasta"
done | awk '{s+=$1} END{print s}'

```

"After splitting the downloaded genomes into target and neighbor sets, I run Fur to search for marker regions. As mentioned above, I could find at least one marker region for 96 of the taxa, with the number of detected marker regions varying widely. Figure 3.21A shows the distribution of the number of markers found for each taxon. On average, 420 marker regions are detected across all organisms, but less than 111 marker regions could be detected for half of them."

Back then rru had 5376 marker regions, now it has 1489 sco now has the highest number of the marker regions which is 3738

the longest marker region was 41.5 kb long and it belonged rba now the longest marker region is 64.3 kb long and within rru

```

for d in */; do
    [ -f "$d/markers.fasta" ] || continue

    awk -v tax="$d" '
    /^>/{
        if(len>max) max=len
        len=0
        next
    }
    {len+=length($0)}
    END{
        if(len>max) max=len
        print tax,max
    }' "$d/markers.fasta"

done | sort -k2 -n | tail -1

```

1.3.8 primers

74 primer pairs

lengths of the primers:

```

grep -h -v '^>' */primers.fasta | awk '{print length}' | sort | uniq -c

```

```
1 17 1 18 4 19 135 20 3 21 1 22 2 24 1 25
```

biggest percentage of the priers have 20 bps

minimum penalty:

```
grep -h "penalty:" */primers.fasta | awk '{print $3}' | sort -n | head -1
```

```
0.000142
```

maximum penalty:

```
grep -h "penalty:" */primers.fasta | awk '{print $3}' | sort -n | tail -1
```

```
9.727346
```

average penalty:

```
grep -h "penalty:" */primers.fasta | awk '{s+=$3} END{print s/NR}'
```

```
0.623132
```

the taxa which have the lowest penalty:

```
for d in */; do
    [ -f "$d/primers.fasta" ] || continue
    echo "$(grep 'penalty:' "$d/primers.fasta" | awk '{print $3}')
```

```
0.000142 ame/ same result as before
```

the taxa which have the highest penalty:

```
for d in */; do
    [ -f "$d/primers.fasta" ] || continue
    echo "$(grep 'penalty:' "$d/primers.fasta" | awk '{print $3}')
```

5.323928 bpa/ 8.964893 bap/ 9.727346 cdi/ bpa was the worst before, now
we have bap and cdi as well

1.3.9 anam.sh

we need to make some changes in the code so that it will continue looping
before:

```
if [[ $assembly == "" ]]; then
    continue
```

after:

```
if [[ $assembly == "" ]]; then
    cd -
    continue
```

i ran the script:

```
bash ../scripts/anam.sh < ../data/list.txt
```

1. Results

Only two taxa have anam.txt

bfa: #Annotation for interval (1329064, 1329155)

CP001643.1 Genbank region 1 3614992 . + . ID=CP001643.1:1..3614992;

Dbxref=taxon:446465;Is_circular=true;Name=ANONYMOUS;culture-collection=NCIMB:9860;gbkey=genome=chromosome;mol_type=genomic DNA;note=type strain of Brachybacterium faecium DSM 4810; strain=DSM 4810;type-material=type strain of Brachybacterium faecium

CP001643.1 Genbank gene 1328670 1329350 . - . ID=gene-Bfae₁₁₇₂₀;

Name=Bfae₁₁₇₂₀;gbkey=Gene;gene_biotype=protein_coding;locus_tag=Bfae₁₁₇₂₀

CP001643.1 Genbank CDS 1328670 1329350 . - 0 ID=cds-ACU85018.1;

Parent=gene-Bfae₁₁₇₂₀;Dbxref=NCBI_{GP}:ACU85018.1;Name=ACU85018.1;gbkey=CDS;locus_tag=product=hypothetical protein;protein_id=ACU85018.1;transl_table=11

hin: #Annotation for interval (1149673, 1149822) L42023.1 Genbank

region 1 1830138 . + . ID=L42023.1:1..1830138; Dbxref=taxon:71421;Is_circular=true;Name=ANON DNA; old-name=Haemophilus influenzae Rd;strain=Rd KW20

L42023.1 Genbank gene 1149653 1150297 . - . ID=gene-HI₁₀₈₄;Name=HI₁₀₈₄;

Note=HI1084;gbkey=Gene;gene_biotype=protein_coding;locus_tag=HI₁₀₈₄

L42023.1 Genbank CDS 1149653 1150297 . - 0 ID=cds-AAC22740.1;

Parent=gene-HI₁₀₈₄;Dbxref=NCBI_{GP}:AAC22740.1;Name=AAC22740.1;

Note=similar to SP:P45390 PID:606130 GB:U00096 PID:1789583 percent identity: 42.03%3B identified by sequence similarity%3B putative;

gbkey=CDS;locus_tag=HI₁₀₈₄;product=conserved hypothetical protein;protein_id=AAC22740.1;tran